Software Engineering Lab 4

PES1UG24CS579 | Actual chat history

Assigned project: SETAPESU26/30_donkeykong

Export cutoff: 07 October 2026, 11:38:32 IST. 18 visible user and assistant messages.

Source: the original local conversation transcript. Message text is reproduced without summaries or reconstructed replies; Markdown syntax is retained. System metadata, tool calls/results, and private reasoning are excluded. This is a snapshot through the cutoff; later messages require a refreshed export.

1. Student | 07 October 2026, 11:16:54 IST

I’m doing my Software Engineering Lab 4 assignment, My SRN is PES1UG24CS579, and my assigned repository is https://github.com/SETAPESU26/30_donkeykong. Please work with the files that have already been set up locally. The original repository is cloned at /Users/zahoormashahir/Documents/ChatGPT/SE/source/30_donkeykong, and my personal lab repository is at /Users/zahoormashahir/Documents/ChatGPT/SE/source/SE-Labs-PES1UG24CS579. The working game is inside the Lab-4 folder of my personal repository. Use /Users/zahoormashahir/Documents/ChatGPT/SE/work/lab4-venv/bin/python to run it; Python 3.12 is already installed there.
Start by reading Lab-4/README.md and game.py, checking the Git status and remotes, My lab requires a 10-second gameplay video before , another one after all changes, updated code, and the chat history exported as a document or PDF. The repository README additionally asks for the shared chat link. Please preserve these requirements throughout our work.
So i have taken the video already, the barrels in the top layer fall down too often making it hard to complete the game, . Find the ladder-descent probability check inside Barrel.update and correct it so each eligible encounter has approximately a 30% chance of descending. Preserve the existing one-decision-per-ladder behaviour and the rest of the game. Check the decision boundary using controlled random values and, if useful, a repeatable larger sample using the actual Barrel logic. Run the game again to check that the fix has not broken normal movement.
Commit this fix separately with a clear message. Explain the cause, the exact change, and what you checked in straightforward language so I can review it. Leave the three feature functions for my next prompts. Keep all assignment changes inside Lab-4 in my personal repository, preserve my earlier labs, and do not push to SETAPESU26 or create a pull request. Wait for my next prompt after finishing this task.

2. Assistant | 07 October 2026, 11:16:57 IST

I’ll read the lab instructions and game code, check both repositories, then make and verify the barrel fix. I’ll keep the assignment edits inside your personal repository’s Lab-4 folder and commit the fix separately.

3. Assistant | 07 October 2026, 11:17:37 IST

The cause is in `Barrel.update`: `random.random() < 0.7` gives each eligible encounter a 70% descent chance. Changing it to `< 0.3` will give the intended 30% chance, while keeping the existing `skip` set that prevents another decision at the same ladder.

I’m using the debugging skill to check the boundary and movement. Lab-4 is currently untracked, so I’ll first commit the supplied files as a baseline, then commit the fix separately. I’ve also noted your recorded before video and the remaining final video, chat export, and shared-link requirements.

4. Assistant | 07 October 2026, 11:18:43 IST

The checks pass: values below `0.3` descend, while `0.3` and above keep rolling. Repeated frames and revisits do not reroll the decision. A seeded sample of 10,000 encounters through the actual `Barrel.update` logic produced 2,988 descents—29.88%.

The movement checks also passed. I'm checking the visible game now; macOS required display access outside the sandbox to open its window.

5. Assistant | 07 October 2026, 11:20:24 IST

The real game completed an 11-second controlled run successfully. Walking, ladder climbing, movement along the slope, jumping and landing, reset, barrel spawning, and rendering all worked.

I'll now record the results and submission requirements in the Lab-4 README, then commit the fix. The three feature functions remain untouched.

6. Assistant | 07 October 2026, 11:21:00 IST

Committed as `c76e135` — **Fix Lab 4 barrel ladder descent probability to 30 percent**.

The cause was a `< 0.7` check, giving a 70% descent chance. [game.py](/Users/zahoormashahir/Documents/ChatGPT/SE/source/SE-Labs-PES1UG24CS579/Lab-4/game.py:167) now uses `< 0.3`. The existing one-decision-per-ladder behavior is preserved.

Verified:

- Below `0.3` descends; exactly `0.3` and above continue rolling.
- Seven checks passed, including repeat encounters and movement.
- 10,000 seeded encounters produced **29.88%** descents.
- An 11-second real game run passed walking, climbing, jumping, reset, and rendering checks.

Lab-4 was first committed as baseline `fbd1217`, keeping the fix separate. Earlier labs and all three feature functions are untouched. Nothing was pushed; no PR was created.

The README records your before video and the remaining final video, chat export, and shared-link requirements. Waiting for your next prompt.

7. Student | 07 October 2026, 11:23:06 IST

Continue with Tasks 2 and 3 from the README. Implement theme_color(score) so the background starts navy, becomes dark purple at 500 points, and changes to a warmer dark colour at 1000 points. Keep everything readable and make sure resetting restores the original background. Check it and commit this feature separately. Then implement on_barrel_jumped(player, barrel) to show a floating points label above the barrel for about one second. It should appear only once per successful jump and display the actual points awarded, so it will still be correct when we add the multiplier. Keep the existing callback interface and clear temporary effects when restarting. Test it in the game and make a separate commit.
Keep the code simple and consistent with the existing game. Explain what you changed and what you verified, then wait for my next prompt.

8. Assistant | 07 October 2026, 11:23:10 IST

I'll implement and verify the score-based background first, then commit it separately. After that I'll add the floating points label, check that it shows the awarded amount once per jump and clears on restart, and make a second commit.

9. Assistant | 07 October 2026, 11:24:19 IST

The background now stays navy below 500 points, turns dark purple at 500, and becomes a warm dark brown at 1000. I checked both thresholds and the rendered scenes; the game elements remain readable. Restarting after a 1000-point win restored navy.

I'll commit that feature now. For the label, I'll store the calculated award on the barrel before calling the existing two-argument callback, so it can show the correct amount when the multiplier is added.

10. Assistant | 07 October 2026, 11:26:05 IST

The floating label passed the game checks: a successful jump showed one `+100` label, and a temporary test multiplier correctly produced `+250`. The label moved upward, disappeared after one second, and cleared when restarting while it was visible.

I also reran the seven barrel and movement checks; all still pass. I'll record these results and make the second feature commit.

11. Assistant | 07 October 2026, 11:26:24 IST

Completed in two separate commits:

- **`7b45a51` — Background colours:** navy below 500 points, dark purple at 500, warm dark brown at 1000. Checked threshold values, readability, and that restarting restores navy.
- **`9d6ef41` — Floating points labels:** one label per cleared barrel, rising for about one second. It displays the exact awarded bonus while preserving `on_barrel_jumped(player, barrel)`. Restarting clears temporary effects.

Real game checks passed for `+100` and a test-only `+250` award, single appearance, upward movement, expiry, and reset. All seven existing barrel/movement checks also passed.

Only Lab-4 changed. Task 4 remains untouched, submission requirements remain recorded in the README, and nothing was pushed. Waiting for your next prompt.

12. Student | 07 October 2026, 11:26:52 IST

Now implement Task 4, score_multiplier(score). Award 1x points below 1000 and 2x at 1000 or above, using the score before the jump bonus is added. Display the multiplier in the HUD and make sure the floating label shows the correct bonus. Reset everything properly when R is pressed, test the threshold, and commit this feature separately.
Help me record a genuine ten-second after video showing the completed game and its features. Save it alongside the before video under Lab-4/videos. Use actual gameplay, and guide me if I need to operate the recorder or play.
Prepare the chat-history PDF from the actual conversation and help me obtain its shared link. If you cannot access the full history, ask me for the export rather than inventing it. Update the Lab-4 README with setup instructions, completed tasks and deliverable locations. Keep only useful submission files and leave everything ready for a final code review before pushing.

13. Assistant | 07 October 2026, 11:26:55 IST

I'll add and verify the multiplier first, then commit it separately. After that I'll check the video files and recording options, prepare the chat-history PDF from the actual conversation, and update the README for final review.

14. Assistant | 07 October 2026, 11:28:41 IST

I found the actual conversation transcript locally, so I can make the PDF without reconstructing any messages. I'm using the PDF skill for the export and OpenAI Docs to check the shared-link process.

For the after video, play until you're near 1000 points, then record a ten-second segment that shows the purple-to-warm background change, the HUD switching to 2x, and a `+200` jump label. I'll help you save and verify the clip. The earlier controlled test runs won't be used as gameplay evidence.

15. Assistant | 07 October 2026, 11:30:22 IST

Task 4 is committed as `07dd1d0`. All ten checks pass, including `999 → 1x`, `1000 → 2x`, a `900 → 1000` jump awarding `+100`, and the next jump awarding `+200`. The HUD shows the multiplier for the next jump. R restores navy, 1x, three lives, clears barrels and labels, and resets the spawn delay.

The completed game is open. Collect points before recording; the clip doesn't need to start from a fresh game. Press **Shift–Command–5**, choose **Record Selected Portion**, frame the game window, and choose `Lab-4/videos` as the save location. Record at least ten seconds of active play, then stop from the menu bar. I can trim and verify the saved recording.

16. Assistant | 07 October 2026, 11:34:25 IST

The PDF currently contains the actual recorded conversation through its stated cutoff. The on-disk transcript hasn't yet included the later messages from this active turn, so it will need a refresh after this turn ends.

The sharing tool hasn't returned a link, and computer control cannot open this app's sharing interface. Please use **Share → Copy link** in this chat and paste the URL here. Shared links are snapshots, so create or refresh it after the assignment work is complete. [OpenAI documentation](https://learn.chatgpt.com/docs/use-chatgpt)

17. Assistant | 07 October 2026, 11:36:39 IST

I found two Donkey Kong recordings on your Desktop from before the fix. The later one, recorded at 11:07 AM, shows active play and barrels taking ladders. I'll use a ten-second segment from that recording and leave the original untouched.

The README will clearly distinguish finished deliverables from the after video and shared link that still need to be supplied.

18. Assistant | 07 October 2026, 11:38:32 IST

The before clip is saved under `Lab-4/videos` and verified as exactly 10.000 seconds. The README now covers setup, all completed tasks, checks, recording guidance, and deliverable locations. I've also queued the game changes for review in the app.

I'll refresh the PDF to the latest available messages and commit the submission files. The after recording and shared URL remain pending; I'll need those to finish the evidence package.